

Food Delivery Time Prediction – README

Project Files Overview

This project compares multiple **Machine Learning (ML)** and **Deep Learning (DL)** models for predicting food delivery time using engineered features derived from delivery, weather, and temporal data.

File Names

◆ Deep Learning Models

- Newdata_DL_model.ipynb / .py
 - Simple Neural Network (Dense NN)
 - LSTM Model
 - Residual Neural Network

◆ Machine Learning Models

- Newdata_ML_model.ipynb / .py
 - Linear Regression
 - Lasso Regression
 - Decision Tree Regressor
 - Random Forest Regressor
 - Gradient Boosting Regressor
 - Extra Trees Regressor
 - XGBoost Regressor (Base)
 - LightGBM Regressor
 - CatBoost Regressor

◆ Fine-Tuned Model (Separate File)

- FineTunned_XgBoost.ipynb
 - Hyperparameter tuned XGBoost model
 - Contains optimized training pipeline and best-performing configuration
- Newdata_ML_model.ipynb / .py

- Linear Regression
 - Lasso Regression
 - Decision Tree Regressor
 - Random Forest Regressor
 - Gradient Boosting Regressor
 - Extra Trees Regressor
 - XGBoost Regressor (Base + Tuned)
 - LightGBM Regressor
 - CatBoost Regressor
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Columns Used in Dataset

◆ Raw Features

- Delivery_person_Age
- Delivery_person_Ratings
- Restaurant_latitude
- Restaurant_longitude
- Delivery_location_latitude
- Delivery_location_longitude
- Vehicle_condition
- multiple_deliveries

◆ Time-Based Features

- Year
- Month
- Day
- DayOfWeek
- IsWeekend
- Order_Hour
- Pickup_Hour

◆ Distance Feature

- Distance_km

◆ Encoded / Engineered Features

- Delivery_person_num
- Age_Rating_Interaction
- Distance_Vehicle_Interaction
- Weatherconditions_Encoded
- Road_traffic_density_Encoded
- Type_of_order_Encoded
- Type_of_vehicle_Encoded
- Festival_Encoded
- City_Encoded
- Delivery_city_Encoded
- Order_Time_Category_Encoded
- Pickup_Time_Category_Encoded
- Age_Group_Encoded
- Rating_Group_Encoded
- Distance_Group_Encoded
- DayName_Encoded

Feature Engineering

The following feature engineering techniques were applied:

1 Temporal Feature Engineering

- Extracted hour, day, month, weekday, and weekend indicators
- Created categorical bins (time categories)

2 Geospatial Engineering

- Calculated Distance_km using latitude and longitude

3 Interaction Features

- Age × Rating → Age_Rating_Interaction
- Distance × Vehicle Condition → Distance_Vehicle_Interaction

4 Encoding Techniques

- Label Encoding for categorical columns
- Group Encoding (Age, Rating, Distance bins)
- City-level encoding for delivery patterns

5 Behavioral Features

- Delivery_person_num (experience proxy)
- Multiple deliveries indicator

Model Performance Summary

◆ Deep Learning Results

Model	R ² Score	RMSE	Performance
Simple NN	-3.59	20.26	✖ Poor (Failed to learn)
Residual NN	-3.59	20.26	✖ Poor
LSTM	0.38	7.40	⚠ Moderate

📌 Observation:

- Feedforward networks performed poorly
- LSTM improved results but still weaker than ML models

◆ Machine Learning Results

Model	Test R ²	RMSE	Notes
Linear Regression	0.49	6.73	Baseline
Lasso	0.49	6.73	Similar to Linear
Decision Tree	0.81	4.10	Good

Model	Test R ² RMSE Notes		
Random Forest	0.82	3.98	Strong
Gradient Boosting	0.80	4.18	Stable
Extra Trees	0.76	4.58	Moderate
XGBoost (Base)	0.82	3.95	Strong
LightGBM	0.83	3.91	Very Strong
CatBoost	0.82	3.94	Strong
 XGBoost (Tuned)	0.83	3.88	 Best Model

Best Model

XGBoost (Fine-Tuned)

Performance:

- R² (Test): **0.8313**
- RMSE: **3.88**
- Lowest prediction error among all models

Why it performed best:

- Handles non-linearity well
 - Robust to feature interactions
 - Effective with engineered tabular data
 - Boosting reduces bias and variance
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Key Insights

- Tree-based ensemble models outperform deep learning on tabular delivery datasets
- Feature engineering contributed more than model complexity
- Temporal and distance-based features are highly influential
- Tuned boosting models consistently give best results

Conclusion

After evaluating multiple Machine Learning and Deep Learning approaches, **tree-based ensemble models significantly outperformed neural networks** for this tabular delivery prediction task.

Among all models tested, the **fine-tuned XGBoost model achieved the best overall performance**, with the highest R^2 score and lowest RMSE. This indicates strong generalization and accurate prediction of delivery time.

Deep learning models, especially basic dense and residual networks, failed to capture patterns effectively, likely due to the structured nature of tabular data. While LSTM showed moderate performance, it still lagged behind boosting models.

Final Recommendation:

Use **Fine-Tuned XGBoost** for deployment due to its superior accuracy, robustness, and scalability on real-world delivery datasets.